Cambridge (CIE) A Level Computer Science

Your notes

Constructs

Contents

- Selection
- Iteration

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Your notes

Selection

What is selection?

- Selection is when the **flow of a program is changed,** depending on a **set of conditions**
- The outcome of this condition will then determine **which lines or block of code is run next**
- Selection is used for **validation**, **calculation** and making sense of a **user's choices**
- There are two ways to write selection statements:
  - **if**... **then**... **else**...
  - **case**...

If statements

What is an if statement?

- An If statements allow you to **execute a set of instructions if a condition is true**
- They have the following syntax:

Pseudocode	

IF <condition> THEN
  <statement(s)>
ENDIF
	IF <condition> THEN
	  <statement(s)>
	ELSE
	  <statement(s)>
	ENDIF

Without an ELSE clause	With an ELSE clause

Nested if statements

- Nested if statements are **an if statement within an if statement**
- Nested means to be '**stored inside the other**'

Example code

```
IF Player2Score > Player1Score THEN
   IF Player2Score > HighScore THEN
      OUTPUT Player2, " is champion and highest scorer"
   ELSE
      OUTPUT Player2, " is the new champion"
   ENDIF
ELSE
   OUTPUT Player1, " is still the champion"
```


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```
  IF Player1Score > HighScore THEN
    OUTPUT Player1, " is also the highest scorer"
  ENDIF
ENDIF
```

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If statements in different languages

Python	VB.net

weather = input("What is the weather like today? (sunny, rainy, snowy): ")
 	
if weather == "sunny":
  print("Don't forget your sunglasses!")
elif weather == "rainy":
  print("Take an umbrella with you.")
elif weather == "snowy":
  print("Wear warm clothes!")
else:
  print("Not sure what to suggest for that kind of weather.")
	Module WeatherCheck
	  Sub Main()
	  Dim weather As String
	
	  Console.Write("What is th
	  weather = Console.ReadL
	
	  If weather = "sunny" Ther
	  Console.WriteLine("Dor
	  ElseIf weather = "rainy" Tl
	  Console.WriteLine("Tak
	  ElseIf weather = "snowy" '
	  Console.WriteLine("We
	  Else
	  Console.WriteLine("Not
	  End If
	
	  Console.ReadLine() ' Keep
	  End Sub
	End Module

Case statements

What is a case statement?

- A case statement **can mean less code** but it only useful when comparing **multiple values of the same variable**
- **If** statements are **more flexible** and are generally used more in languages such as Python
- The format of a CASE statement is:

Pseudocode

CASE of <identifier>
  <value 1> : <statement1>
  <statement2>
  ...
  <value 2> : <statement1>
  <statement2>
  ...
  ...
ENDCASE
	CASE of <identifier>
	  <value 1> : <statement1>
	  <statement2>
	  ...
	  <value 2> : <statement1>
	  <statement2>
	  ...
	  OTHERWISE : <statement1>
	  <statement2>
	  ...
	ENDCASE

	An OTHERWISE clause can be the last clause

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Example code

```
DECLARE Direction : STRING

OUTPUT "Enter a direction (N, S, E, W):"
INPUT Direction

CASE OF Direction
   "N" : OUTPUT "You are heading North"
   "S" : OUTPUT "You are heading South"
   "E" : OUTPUT "You are heading East"
   "W" : OUTPUT "You are heading West"
   OTHERWISE : OUTPUT "Invalid direction entered"
ENDCASE
```

Case statements in different languages

Python	VB.net

weather = input("What is the weather like today? (sunny, rainy, snowy): ")
 	
match weather:
  case "sunny":
  print("Don't forget your sunglasses!")
  case "rainy":
  print("Take an umbrella with you.")
  case "snowy":
  print("Wear warm clothes!")
  case _:
  print("Not sure what to suggest for that kind of weather.")
	Module WeatherCheck
	  Sub Main()
	  Dim weather As String
	
	  Console.Write("What is th
	  weather = Console.ReadL
	
	  Select Case weather
	  Case "sunny"
	  Console.WriteLine("D
	  Case "rainy"
	  Console.WriteLine("T
	  Case "snowy"
	  Console.WriteLine("W
	  Case Else
	  Console.WriteLine("N
	  End Select
	
	  Console.ReadLine() ' Keep
	  End Sub
	End Module

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Iteration

Your notes

What is iteration?

- Iteration is **repeating a line or a block of code** using a loop
- Iteration can be:
  - **Count-controlled**
  - **Condition-controlled**

Count-controlled loops

What is a count-controlled loop?

- A count-controlled loop is when the code is repeated **a fixed number of times** (e.g. using a FOR loop)
- A count-controlled loop can be written as:

Pseudocode	
FOR <identifier> ← <value1> TO <value2> <statements(s)> NEXT <identifier>	FOR <identifier> ← <value1> TO <value2> STEP <increment> <statements(s)> NEXT <identifier>

- In a **FOR loop**, the **increment** must be an **expression that evaluates to an integer**
  - The loop **starts** at value1
  - The identifier is updated by the **increment** value on each iteration
  - The loop continues until the identifier **passes value2**
- The increment can be **positive** or **negative**, depending on the direction you want the loop to count

Nested count-controlled loops

```
FOR i ← 1 TO 3
  FOR j ← 1 TO 3
    OUTPUT i, " x ", j, " = ", i * j
  NEXT j
NEXT i
```

- The **outer loop** runs i from 1 to 3
- For **each value of** i, the **inner loop** runs j from 1 to 3
- It outputs all combinations of i × j

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
```


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2 x 1 = 2
...
3 x 3 = 9

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Count-controlled loops in different languages

Python	VB.net	Java
Print the numbers 1 to 5		

for i in range(1, 6):
print(i)
	For i As Integer = 1 To 5
	Console.WriteLine(i)
	Next
		for (int i = 1; i <= 5; i++) {
		System.out.println(i);
		}

Condition-controlled loops

What is a condition controlled loop?

- A condition controlled loop is when the code is repeated **until a condition is met**
- There are two types of condition controlled loops:
  - **Post-condition (REPEAT)**
  - **Pre-condition (WHILE)**

Post-condition loops (REPEAT)

- A post-condition loop **is executed at least once**
- The condition must be an **expression** that evaluates to a **Boolean** (True/False)
- The condition is tested **after the statements are executed** and only stops once the condition is evaluated to True
- It can be written as:

Pseudocode
REPEAT <statement(s)> UNTIL <condition>

Pre-condition loops (WHILE)

- The condition must be an **expression** that evaluates to a **Boolean** (True/False)
- The condition is tested and statements are **only executed if the condition evaluates to True**
- After statements have been executed the condition is **tested again**
- The loop ends when the condition **evaluates to False**

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- It can be written as:

Pseudocode

WHILE <condition> DO
  <statement(s)>
ENDWHILE

Nested condition-controlled loops

```
DECLARE Username : STRING
DECLARE Count : INTEGER

Count ← 0

WHILE Count < 3
   Username ← ""

   WHILE Username = ""
      OUTPUT "Enter a username: "
      INPUT Username

      IF Username = "" THEN
         OUTPUT "Username cannot be blank."
      ENDIF
   ENDWHILE

   OUTPUT "Username accepted: ", Username
   Count ← Count + 1
ENDWHILE
```

- Ask the user to **enter up to 3 usernames**
- For each one, keep asking **until a non-empty name** is entered
- **Outer WHILE** runs 3 times (for 3 usernames)
- **Inner WHILE** ensures that each username is **not blank**

Condition-controlled loops in different languages

Python	VB.net	Java
Keep asking the user to enter a password until they type "ac		

`password = ""`
 	
`while password != "admin":`
`  password = input("Enter the password: ")`
 	
`  if password != "admin":`
`  print("Incorrect, try again.")`
 	
`print("Access granted!")`
	`Module PasswordCheck`
	`  Sub Main()`
	`  Dim password As String = ""`
	
	`  While password <> "admin"`
	`  Console.Write("Enter the password: ")`
	`  password = Console.ReadLine()`
	
	`  If password <> "admin" Then`
	`  Console.WriteLine("Incorrect, try again.")`
	`  End If`
		`import`
		
		`public c`
		`  publi`
		`  Sca`
		`  Str`
		
		`  wh`
		`  S`
		`  p`

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```
    End While

    Console.WriteLine("Access granted!")
    Console.ReadLine() ' Keeps the console open
  End Sub
End Module
```

```
        i

        }
    }

    Sys
    sca
  }
}
```

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When to use each type of loop

Loop type	When to use	Example scenario
Count-controlled loop (FOR)	When you know **in advance** how many times you want the loop to run	Repeating an action 5 times, generating a multiplication table
Pre-condition loop (WHILE)	When you want to **check a condition before** running the loop. It may run **0 or more** times	Keep asking for a valid password before granting access
Post-condition loop (REPEAT ... UNTIL)	When you want the loop to run **at least once**, and then stop when a condition becomes true	Ask the user for input and validate it after first run